

Tide retrospective: *the fourth connected cumulant (flow-equation identity)*

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Abstract

The capstone of the anharmonic cumulant arc: the fourth connected cumulant of $u = -(tx)$ as the third derivative of the perturbed mean at $h = 0$,

$$\kappa_4 = \partial_h^3|_0 \langle u \rangle_h = \langle u^4 \rangle_0 - 4\langle u^3 \rangle_0 \langle u \rangle_0 - 3\langle u^2 \rangle_0^2 + 12\langle u^2 \rangle_0 \langle u \rangle_0^2 - 6\langle u \rangle_0^4,$$

in closed $(G_0, \dots, G_4)$ form. Three derivative layers — mean $M = G_1/G_0$, susceptibility $S = \text{deriv } M$, third cumulant $S_2 = \text{deriv } S$ — stitched with `Filter.EventuallyEq.deriv`. New file `AnharmonicFourthCumulant.lean`; builds within the default heartbeat budget, no `sorry/axiom/native_decide`.

1 Setting

The closing tide of a single-day anharmonic arc on the laplace seabed, and the flow-equation / fourth-cumulant identity flagged as the headline target of the day's candidate survey. The cumulant ladder was built rung by rung: the mean (K') , the susceptibility κ_2 (∂_h of the mean), its general- h version, and κ_3 (∂_h^2 of the mean). κ_4 is ∂_h^3 of the mean — one differentiation layer beyond κ_3 , which is what made it the heaviest single tide.

2 The thing formalised

Writing G_n for `weightedPartition lam alpha gamma t n`:

For the anharmonic potential, $\lambda, \gamma > 0$, $\alpha^2 < 3\lambda\gamma$, $t > 0$,

$$\begin{aligned} & \text{iteratedDeriv } 3 \ (h \mapsto G \ 1 \ h / G \ 0 \ h) \ 0 \\ &= \frac{G_4 G_0^3 - 4G_3 G_1 G_0^2 - 3G_2^2 G_0^2 + 12G_2 G_1^2 G_0 - 6G_1^4}{G_0^4}, \end{aligned}$$

all G_n at 0 — the standard moments-to-cumulants combination after dividing each $G_n(0)$ by $G_0(0)$.

A companion theorem, `anharmonic_susceptibility_deriv_general`, gives the third cumulant at any $|h_0| < 1$ (the general- h `deriv S`), which is what the chaining consumes.

3 Proof strategy

Three layers. The mean's derivative is the susceptibility S (general- h susceptibility tide); S 's derivative is $S_2 = (G_3 G_0^2 - 3G_1 G_2 G_0 + 2G_1^3)/G_0^3$ (proved here at general h_0 by a compound quotient rule); and S_2 's derivative at 0 is κ_4 (a second compound quotient rule, the numerator built from `.mul/.pow/.const_mul/.sub/.add`). The three are joined: `deriv M =ev S` and `deriv S =ev S2` on `ball 0 1`, so by `Filter.EventuallyEq.deriv` the second derivatives agree eventually too, and `iteratedDeriv 3 M 0 = deriv(deriv(deriv M)) 0 = deriv S2 0 =  $\kappa_4$` .

4 Roadblocks and resolutions

Two, both mechanical. (i) *set_option placement*. The `maxHeartbeats` bumps I added pre-emptively sat between `docstring` and `theorem`, which is a parse error (the `docstring` must about the declaration); they turned out to be unnecessary anyway — the degree-eight `ring` closes within the default budget — so they were simply removed. (ii) *Pi-vs-pointwise*. Building the numerator's `HasDerivAt` from sub-terms like `hG0.pow 2` leaves `Pi.pow/Pi.mul applications ((G02) 0` rather than `(G0 0)2` inside the derivative value, which `ring` treats as opaque atoms distinct from their pointwise forms. An ascription of the top-level function fixes that function but not the value; a `simp only [Pi.pow_apply, Pi.mul_apply]` before `field_simp` pushes the applications inward and lets `ring` finish. (The third-cumulant tide avoided this only because its sub-terms happened to stay pointwise.)

5 What was Mathlib, what was new

Mathlib. The `HasDerivAt` combinators (`.mul`, `.sub`, `.add`, `.pow`, `.const_mul`, `.div`, `.deriv`); `iteratedDeriv_succ/_one`; `Filter.EventuallyEq.deriv`; the `Pi._apply simp` lemmas; `field.simp`, `ring`.

Seabed (reused). The pointwise partition derivatives, the general- h susceptibility, and the $G_0(0) > 0$ positivity — all from this session's arc.

New. The general- h third cumulant and the fourth-cumulant theorem. About 160 lines.

6 Lessons

- **Ascribe the *sub-terms*, not just the outer function.** A function ascription on a compound `HasDerivAt` does not stop `Pi.pow/Pi.mul` from appearing in the inferred derivative *value*; a `simp only [Pi._apply]` before `ring` is the reliable cleanup for nested quotient/product rules.
- **Try the default heartbeat budget before bumping it.** Even the degree-eight cumulant `ring` fit the default; the pre-emptive `maxHeartbeats` was dead weight (and triggers a style linter).
- **The cumulant ladder is uniform.** $\kappa_2, \kappa_3, \kappa_4$ are the same recipe at increasing order: differentiate the previous cumulant function (needing its general- h form) and chain. κ_5 would be one more identical layer.

7 Follow-ups

- **Connect to $\log Z$ explicitly.** Package $\kappa_2, \dots, \kappa_4$ as $\partial_h^n \log Z|_0$ via the cumulant generating function, closing the loop with `logPartition_hasDerivAt`.
- **Moment-form corollaries.** State each κ_n in the $\langle u^k \rangle_0$ (`gibbsExp`) form via moment-normalisation.
- **Promote `amgm_t_abs_x` to `lean-common`.**